

LARAIB KHAN

BEEL- COMPUTER SYSTEMS

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EDUCATION

Bachelors of Electrical Engineering Oct 2022 - July 2026
Major in Computer Systems Engineering, Uit University

Intermediate: May 2020 - August 2022
Pre Engineering , Govt Degree College Majeed SRE |||

PROFESSIONAL EXPERIENCE

IC Design and Verification Trainee at UIT University May 2026 - Current

- Receiving hands-on training in Digital IC Design, Verilog HDL, RTL Design, and FPGA implementation.
- Designed and implemented digital hardware projects, including a UART communication system.
- Gaining practical experience in digital logic design, FSM implementation, simulation, and hardware validation.
- Building a strong foundation in semiconductor design methodologies and hardware development.

PROMPT ENGINEERING INTERN at Exelerate IT (Powered by RIT) Aug 2024 - Sep 2024

- Worked with Generative AI and Large Language Models (LLMs) to develop and optimize prompts.
- Conducted prompt testing and evaluated AI-generated responses for improved accuracy.
- Collaborated on AI-driven projects and explored real-world applications of prompt engineering.
- Enhanced problem-solving, research, and communication skills through practical tasks.

Electrical Engineering Intern at Pakistan International Airlines (PIA) Aug 2024 - Sep 2024

- Assisted in developing marketing campaigns and promotional materials.
- Coordinated content updates across digital platforms.
- Conducted market research and tracked campaign performance.
- Assisted in preparing marketing reports and campaign insights.

CERTIFICATES

- Introduction to Artificial Intelligence – Coursera (Andrew Ng)
- Electrical Engineering Internship Certificate – Pakistan International Airlines (PIA)
- Prompt Engineering – Exelerate IT (Powered by RIT)

TECHNOLOGIES

Programming Languages:

- C++ , Python , Verilog HDL, System Verilog

Technologies:

- RTL Design, FPGA Design, ASIC Design Flow, Physical Design, OpenLane 2.0, Xilinx Vivado, QuestaSim, Static Timing Analysis (STA), APB Protocol, Ubuntu Linux, Git, logisim

DIGITAL IC DESIGN & VERIFICATION PROJECTS

UART Transmitter and Receiver Design:

- Designed and implemented a UART (Universal Asynchronous Receiver/Transmitter) communication system using Verilog HDL on an FPGA platform.
- Developed the UART Transmitter and Receiver modules for reliable serial data communication.
- Implemented Finite State Machine (FSM) based control logic for data transmission and reception.
- Verified the design through simulation and successfully validated the functionality on FPGA hardware.

AMBA Bus Protocol Implementation (APB, AHB & AXI):

- Designed and implemented AMBA bus protocols (APB, AHB, and AXI) to understand on-chip communication between master and slave devices.
- Developed and simulated master-slave communication using control and data transfer mechanisms.
- Implemented address decoding, read/write operations, and protocol-specific control signals.
- Verified data transactions through simulation and analyzed communication flow for different bus architectures.

Cache Controller Design:

- Implemented a Cache Controller to manage memory read/write operations.
- Verified cache functionality using custom Verilog testbenches.

Single-Port SRAM Controller Design:

- Designed a Single-Port SRAM Controller supporting read and write operations.
- Developed and verified the controller using Verilog HDL and simulation tools.

On-Chip Communication (Handshaking Protocol):

- Designed and implemented a single master-single slave communication system using the Handshaking Protocol in Logisim Evolution.
- Developed the communication logic to enable reliable data transfer between the master and slave modules.
- Implemented control signals for synchronization and proper data exchange.
- Verified the design by simulating different communication scenarios and validating data transfer.

ASIC Physical Design using OpenLane 2.0:

- Executed the complete RTL-to-GDSII flow using OpenLane 2.0.
- Performed synthesis, floorplanning, PDN generation, IO placement, placement, CTS, routing, STA, DRC, LVS, and GDSII generation.

ACADEMIC PROJECTS

Cache Simulaor:

- Developed a Cache Memory Simulator using Object-Oriented Programming (OOP).to simulate cache operations and memory access behavior.
- Implemented cache read and write operations using configurable cache parameters.
- Simulated cache performance by tracking cache hits, cache misses, and hit ratio.
- Applied concepts of Computer Architecture, memory hierarchy, and cache management.
- Tested and validated the simulator using different memory access patterns.

AI Based Food Recognition and Nutritional Estimation Plate:

- Developed an AI-powered food recognition system to identify food items and estimate their nutritional values using computer vision techniques.
- Trained and integrated a YOLOv8 model for food detection and classification from real-time camera input.
- Implemented image preprocessing and object detection to improve recognition accuracy.
- Displayed estimated nutritional information, including calories, carbohydrates, proteins, and fats, through a user-friendly interface.
- Gained practical experience in Artificial Intelligence, Computer Vision, Machine Learning, image processing, and real-time system development.

TECHNICAL SKILLS

- Python (Basic)
- Verilog HDL
- Digital Logic Design
- FPGA
- Computer Architecture
- Data Structures
- Prompt Engineering

SOFT SKILLS

- Problem Solving
- Analytical Thinking
- Team Collaboration
- Communication
- Time Management
- Adaptability